



Initiation Setting and Persistence of Medications Affecting Cognition in Older Adults

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Introduction

Medications that adversely affect cognition, eg, antipsychotics, may increase delirium and fall risk among older adults,^{1,2} but they remain frequently prescribed.³⁻⁵ To better understand the context of their prescribing, this study examined the clinical settings in which these medications were initiated and the rate at which patients continue to take them 1 year later.

+ Supplemental content

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Methods

This cross-sectional study used Health and Retirement Study (HRS) data of community-dwelling adults aged 66 years or older linked to Medicare fee-for-service claims (including Part D prescription fills) between January 1, 2008, and December 31, 2021. The University of California, Los Angeles institutional review board approved the study with a waiver of informed consent as the data were deidentified. The study followed the [STROBE](#) reporting guideline for cross-sectional studies.

We studied 4 medication classes from the Beers Criteria¹: benzodiazepines, nonbenzodiazepine hypnotics, antipsychotics, and anticholinergics (eTable in [Supplement 1](#)). We included patients with at least 2 consecutive years of continuous enrollment (to allow for a 1-year lookback period) in Medicare Parts A, B, and D and with no prescription of a given class in the year before the first prescription fill to identify their first initiation in that class. Using HRS's validated cognitive testing data, patients were categorized as having no cognitive impairment; cognitive impairment, not dementia (CIND); or dementia.⁶ We assumed that a prescription was initiated from the patient's last clinical setting (office, emergency department, inpatient, or skilled nursing facility) before the first prescription.

In unadjusted analyses, we first examined the clinical setting for prescription initiation, then compared the percentage of initiations from acute or postacute settings (emergency department, inpatient, skilled nursing facility) with the percentage of overall patient visits to such settings. In covariate-adjusted analyses among the subsample (82.0% of the main sample) of patients continuously enrolled the year following initiation, we examined the binary outcome of whether a patient continued to take a medication in the same class 1 year after initiation.

Adjusted means controlled for age, sex, self-selected race and ethnicity as defined in the HRS (Black or African American, Hispanic [of any race], White, and other), socioeconomic indicator for Medicare-Medicaid dual eligibility or Part D low-income subsidy, and year of initiation. Race and ethnicity were included as important confounders in the adjusted model.

We used survey weights to generate nationally representative estimates. Analyses were performed from April 16, 2025, to March 27, 2026, using Stata, version 18.5 (StataCorp LLC). A 2-sided $P < .05$ indicated statistical significance.

Results

We identified 4955 new medication initiations, representing 23.4 million first prescriptions among Medicare beneficiaries during 2009 to 2021 (mean [95% CI] age, 75.5 [75.5-75.5] years; 33.7% men

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and 66.3% women; 5.4% identifying as Black, 6.2% as Hispanic, 86.4% as White, and 2.1% as other race and ethnicity).

Compared with the overall proportion of patient visits by clinical setting, patients across all cognitive statuses were disproportionately more likely to be in an acute or postacute setting before medication initiation (**Table 1**). For example, for patients with dementia initiating an antipsychotic, 22.0% (95% CI, 21.9%-22.1%) of their overall visits were to acute or postacute (vs office) settings, but 43.3% (95% CI, 43.2%-43.4%) were most recently in an acute or postacute setting when initiating an antipsychotic (*P* for difference < .001) (covariate-adjusted percentages of initiation are shown in **Table 2**). In covariate-adjusted analyses, a high proportion of patients initiating these medications

Table 1. Unadjusted Percentages of Medication Initiation and Overall Visits by Clinical Setting^a

	Medication initiation, % (95% CI)					Overall visits to the acute or postacute setting, % (95% CI) ^c	P value ^d
Medication class ^b	Office setting	Emergency department	Inpatient	Skilled nursing facility	Acute or postacute setting ^c		
Benzodiazepines							
No cognitive impairment	90.0 (90.0-90.1)	5.4 (5.4-5.4)	3.7 (3.7-3.7)	0.9 (0.9-0.9)	10.0 (9.9-10.0)	7.1 (7.1-7.1)	.001
CIND	78.1 (78.0-78.2)	11.6 (11.5-11.6)	6.9 (6.9-7.0)	3.4 (3.4-3.4)	21.9 (21.8-22.0)	11.3 (11.3-11.4)	.006
Dementia	75.8 (75.7-76.0)	7.9 (7.8-8.0)	8.8 (8.7-8.9)	7.5 (7.4-7.6)	24.2 (24.0-24.3)	19.3 (19.2-19.3)	.08
Nonbenzodiazepine hypnotics							
No cognitive impairment	90.1 (90.0-90.1)	2.5 (2.5-2.5)	4.5 (4.4-4.5)	3.0 (3.0-3.0)	9.9 (9.9-10.0)	6.5 (6.4-6.5)	<.001
CIND	83.9 (83.8-84.0)	2.9 (2.9-3.0)	7.7 (7.6-7.7)	5.5 (5.5-5.6)	16.1 (16.0-16.2)	11.3 (11.3-11.3)	.03
Dementia	70.7 (70.5-70.8)	9.8 (9.7-9.9)	11.6 (11.5-11.7)	8.0 (7.9-8.1)	29.3 (29.2-29.5)	18.8 (18.7-18.8)	.02
Antipsychotics							
No cognitive impairment	75.3 (75.2-75.4)	6.2 (6.2-6.3)	10.4 (10.3-10.4)	8.1 (8.0-8.1)	24.7 (24.6-24.8)	13.0 (13.0-13.1)	<.001
CIND	69.2 (69.1-69.4)	7.0 (7.0-7.1)	12.7 (12.6-12.8)	11.1 (11.0-11.2)	30.8 (30.6-30.9)	13.4 (13.4-13.5)	<.001
Dementia	56.7 (56.6-56.8)	11.1 (11.0-11.2)	14.5 (14.4-14.6)	17.7 (17.6-17.8)	43.3 (43.2-43.4)	22.0 (21.9-22.1)	<.001
Anticholinergics							
No cognitive impairment	85.5 (85.5-85.6)	10.3 (10.3-10.3)	3.4 (3.4-3.4)	0.7 (0.7-0.7)	14.5 (14.4-14.5)	6.5 (6.5-6.5)	<.001
CIND	82.8 (82.8-82.9)	13.4 (13.3-13.5)	2.7 (2.7-2.7)	1.1 (1.1-1.1)	17.2 (17.1-17.2)	9.5 (9.5-9.5)	<.001
Dementia	76.1 (76.0-76.2)	14.3 (14.2-14.4)	5.2 (5.1-5.2)	4.5 (4.4-4.5)	23.9 (23.8-24.0)	14.8 (14.8-14.9)	.001

Abbreviation: CIND, cognitive impairment, not dementia.

^a Calculated using nationally representative Health and Retirement Study survey data linked to Medicare fee-for-service claims from 2008 to 2021.

^b The complete list of medications analyzed is provided in the eTable in [Supplement 1](#).

^c Acute or postacute setting refers to emergency department, inpatient, or skilled nursing facility.

^d For the difference between initiation from and overall visits to acute or postacute care settings.

Table 2. Adjusted Percentages of Initiation and Persistence of Medications Affecting Cognition, by Cognitive Status

Variable	Adjusted percentage (95% CI) ^a			
	Benzodiazepines	Nonbenzodiazepine hypnotics	Antipsychotics	Anticholinergics
Prescribed from acute or postacute setting (vs office setting)				
No cognitive impairment	11.4 (9.1-13.7)	11.0 (8.5-13.6)	23.8 (17.0-30.6)	15.7 (13.6-17.8)
CIND	20.6 (14.0-27.3)	13.8 (8.5-19.0)	26.8 (17.4-36.2)	14.6 (10.3-18.9)
Dementia	20.0 (12.1-27.8)	24.2 (14.0-34.5)	36.9 (27.7-46.1)	19.1 (11.7-26.6)
Continuing the following year				
No cognitive impairment	43.4 (39.6-47.3)	48.2 (43.4-53.0)	52.5 (42.8-62.3)	28.1 (25.2-31.1)
CIND	55.8 (47.6-64.0)	39.0 (30.6-47.4)	62.4 (48.5-76.2)	35.5 (29.2-41.8)
Dementia	53.7 (42.6-64.9)	56.6 (43.2-70.0)	67.1 (54.9-79.3)	44.6 (34.5-54.8)

Abbreviation: CIND, cognitive impairment, no dementia.

^a Calculated using nationally representative Health and Retirement Study survey data linked to Medicare fee-for-service claims from 2008 to 2021. The multivariable linear regression underlying the results (which were largely unchanged when using logistic regression) is the binary outcome of (1) the patient's most recent clinical setting before the first prescription being the acute or postacute setting (emergency department, inpatient hospitalization, skilled nursing facility) (coded as 1) vs an office visit (coded as 0) or (2) whether a patient who initiated a class of medication affecting cognition

continued to take a medication in that class 1 year later as a function of age, sex, race and ethnicity, either Medicare-Medicaid dual eligibility or Part D low-income subsidy, year of medication initiation, cognitive status (no cognitive impairment, CIND, dementia), class of medication initiated (benzodiazepines, nonbenzodiazepine hypnotics, antipsychotics, anticholinergics), and the interaction between cognitive status and class of medication. Each observation is a unique combination of patient and class of medication initiated (on the date of the first prescription in that class), and SEs were clustered at the patient level.

continued to take them 1 year later (Table 2); for example, 67.1% (95% CI, 54.9%-79.3%) of patients with dementia initiating an antipsychotic continued to take a medication in the same class 1 year later.

Across all medication classes, 14.2% of patients with no cognitive impairment, 17.2% with CIND, and 21.9% with dementia who initiated a new medication affecting cognition were prescribed from an acute or postacute care setting. Similarly, 38.2% of patients with no cognitive impairment, 44.1% with CIND, and 51.2% with dementia who initiated a new medication continued to take a medication in the same class 1 year later.

Discussion

This cross-sectional study found a disproportionately high percentage of older adults initiating medications that affect cognition from an acute or postacute setting compared with their percentage of visits to such settings. For example, the percentage of antipsychotics initiated from these settings for patients with dementia was double these patients' proportion of visits to these settings. We also found that patients often continued to take medications affecting cognition 1 year later. Study limitations included the assumption that the last setting before medication initiation was where the prescription originated. Our findings suggest that efforts to reduce prescriptions of medications affecting cognition might have the greatest impact targeting prescriptions from acute or postacute settings.

ARTICLE INFORMATION

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Concept and design: Ly, Leng, Mafi.

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SUPPLEMENT 1.

eTable. Medications in Each Class Analyzed

SUPPLEMENT 2.

Data Sharing Statement